University of North Florida

School of Computing

COP3503

# Project 1 – Name Statistics

## Objective:

Apply basic programming principles such as sequence, selection, iteration, and methods in Java.

## Submission:

Project 1 requires that you submit 1 zip file to Canvas as specified below.

- Submit your project via the Canvas course
  - Upload a zip file containing the 4 files required for the project. Follow the naming convention below.
    - Pseudo_code.jpg/png
    - Flowchart.jpg/png
    - Project1.java
    - Debugging.jpg/png
  - The zip file should be named Project1_N#.zip
    - For example: Project1_n00123456.zip
  - The zip file should not contain any additional files
  - The files should be named as shown above
    - Only the zip file name should contain your N#
  - After submitting to Canvas download and double check submission
    - No late files will be taken after the submission date

## Description:

The name statistics program will prompt the user to input a list of names delimited by commas. Then the program will display a list of options for the user to pick from. The program will continue to run and allow the user to select multiple options only exiting when option 0 is selected.

Options:

1: Display List Ordered
2: Display Full Names
3: Display Single Names
4: Display Name Statistics
5: Display Names with Even Length
6: Display Names with Odd Length
7: Display Names not Capitalized
8: Display Most Frequent Name
9: Enter new list of Names
0: Quit Program

Note: Each option should have at least 1 method associated with it. Some of those methods like the one for displaying statistics should have supporting methods. This excludes option 0 “Quit Program”.

# Requirements:

## Documentation Requirements: 30 points

- 10 points: Write pseudo code that outlines the process of option 9 "Display Most Frequent Name"
    - Write the pseudo code BEFORE implementing the code
    - This can be handwritten or typed
    - Make sure you understand the logic of how this can be implemented
- 10 points: Create a flowchart that outlines the menu logic and which methods are called when an option is selected
    - You must use a flowcharting program to create this diagram – NO Hand drawn diagrams
        - Free Recommended option draw.io
    - Flowchart file must be exported from the flowchart program – DO NOT take a Screenshot
    - Create the flowchart after completing the code for Project 1. The diagram must accurately reflect your implementation of project 1
    - You do not need to show the logic for each method
- 10 points: Debugging Screenshot
    - Using an IDE such as eclipse place a breakpoint at one of the print statements for the Display Most Frequent Name option
    - Then run the program in debug mode and enter a list of names selecting option 8
    - At this point the program execution should pause, and the interface should switch into debug mode
    - Locate the variable explorer then take a screenshot that includes the variable explorer and the console output showing the program execution is currently paused

## Minimum Implementation Requirements:-100 points

The requirements listed below must be implemented correctly or no points will be given for the project.

- The program must display the following list of options in the given order for the user to pick from

  1: Display List Ordered
  2: Display Full Names
  3: Display Single Names
  4: Display Name Statistics
  5: Display Names with Even Length
  6: Display Names with Odd Length
  7: Display Names not Capitalized
  8: Display Most Frequent Name
  9: Enter new list of Names
  0: Quit Program

- The program must use the values 1-9 and 0 for the option selection. DO NOT use any other values
- Each option excluding the "Quit Program" option 0 must have at least 1 supporting method
- The program must continue to run until the user selects the "Quit Program" option
- The program must read in a single line of names delimited by commas
    - The program cannot prompt the user for the size of the list
- Do not declare any global/class member variables
    - Global/class member variables are any variable declared outside of a method body

Failure to meet the requirements listed above will result in a zero for the project.

# Style Requirements: Required

## Javadoc's: -10 points

- Each class and method declaration must have a correct Javadoc comment placed above it
    - Add your Javadoc comments last
    - Generate the Javadoc's using an IDE and verify no errors or warning are produced

## Naming Conventions: -10 points

- Each class, variable and method must be named appropriately
    - Variables and methods always start with a lowercase letter
    - Classes always start with an uppercase letter

## Implementation Requirements: 70 points

### Option 1: Display List Ordered: 5 points

- Display the list in alphabetical order
- The Arrays class has a built-in static method called sort
    - The method sorts the array by the Unicode/ASCII value
        - Which means words with capital letters will always come first
    - Using Overloading there is a version of the sort method that takes in a second parameter
        - The second parameter requires a Comparator object which specifies how to sort the array
            - https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/util/Arrays.html#sort(T%5B%5D,java.util.Comparator)
        - The String class provides a Comparator object which specifies to sort Strings ignoring case
            - https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/lang/String.html#CASE_INSENSITIVE_ORDER
- Do not create your own logic for sorting the array – Explore the built-in classes and methods already available
    - If you decided to use an ArrayList the Collections class has a similar sort method

### Option 2: Display Full Names: 3 points

- Display only the full names given

- Full names will always contain at least 1 space
    - The String class provides several instance methods designed to make working with strings easier
        - https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/lang/String.html#method-summary
            - Select the "Instance Methods" tab and look over each method provided

## Option 3: Display Single Names: 2 points

- Display only the single names given
- Single names will never contain spaces
    - The same logic used for option 2 can be used for option 3
        - The not operator might be useful

## Option 4: Display Name Statistics: 25 points

- Display the number of names given: 2 points
    - Full names count as one name
- Display the total letter count for the given input: 3 points
    - Spaces and Commas should not be included in this count
- Display the average length for the given names: 5 points
    - Spaces should not be counted when determining length
    - Round the number to the second decimal point
- Display the shortest name: 3 points
    - Spaces should not be counted when determining length
    - You are guaranteed that one name will always be shorter than the others
- Display the longest name: 2 points
    - Spaces should not be counted when determining length
    - You are guaranteed that one name will always be longer than the others
- Display the population standard deviation of the name lengths: 10 points
    - Spaces should not be counted when determining length
    - Calculator/Formula Link
        - https://www.calculatorsoup.com/calculators/statistics/standard-deviation-calculator.php
    - Make sure to calculate population standard deviation not sample
    - Round the number to the second decimal point
    - The Math class provides several static methods that are useful when implementing formulas
        - https://docs.oracle.com/en/java/javase/17/docs/api/java.base/java/lang/Math.html#method-summary

## Option 5: Display Names with Even Length: 3 points

- Display names with an even length
    - Spaces should not be counted when determining length

## Option 6: Display Names with Odd Length: 2 points

- Display names with an odd length

- Spaces should not be counted when determining length

### Option 7: Display Names not Capitalized: 5 points

- Display any name given that is not capitalized
  - All names should be considered including the individual names that make up the full names
    - All names given in a full name will be space delimited

### Option 8: Display Most Frequent Name: 10 points

- Determine which name occurs the most in the given list
- Do not consider case when comparing names
  - For example, if the list is "Joe,Noah Smith,joe,Kim,Joe,Joe Wright" then Joe occurs 3 times
    - The full name "Joe Wright" would not match the single name "joe"
- If no names occur more than once then print "No Most Frequent Name"
- You are guaranteed that one name will occur more than the other names unless all names occur once

### Option 9: Enter new list of Names: 15 points

- This option should allow the user to enter a new list of names that replaces the current list
- Create a method that is self-contained and handles prompting the user, reading the list of names in, and converting it to the needed format
  - Keep in mind that you only want to open 1 Scanner object on the System.in stream at a time and once it is closed you cannot open it again
- Use this method to read in the list at the beginning of the program
  - DO NOT have duplicate code for reading in the list

### Option 0: Quit Program: Required

- This option should allow the user to exit the program
  - If this option is not selected the program should continue to execute

## Test Cases:

The submitted program must pass all 3 test cases given below. If an option fails any of the test cases no points will be given for that option. When developing and testing your program use the input given below.

Note: Excluding option 1 the order of the output does not need to be exact however the correct and full output must be displayed

## Test Case 1:

### Input:

```
Emily Johnson,michael,Isabella Martinez,aiden,ophia nguyen,Lucas,olivia
kim,Ethan,ava patel,Noah
```

Option 1 Output:

```
[aiden, ava patel, Emily Johnson, Ethan, Isabella Martinez, Lucas, michael,
Noah, olivia kim, ophia Nguyen]
```

Option 2 Output:

```
ava patel
Emily Johnson
Isabella Martinez
olivia kim
ophia nguyen
```

Option 3 Output:

```
aiden
Ethan
Lucas
michael
Noah
```

Option 4 Output:

```
Name Count: 10
Letter Count Total: 82
Avg Name Length: 8.2
Shortest Name: Noah
Longest Name: Isabella Martinez
Population Standard Deviation: 3.66
```

Option 5 Output:

```
ava patel
Emily Johnson
Isabella Martinez
Noah
```

Option 6 Output:

```
aiden
Ethan
Lucas
michael
olivia kim
ophia nguyen
```

Option 7 Output:

```
aiden
ava
patel
```

```
michael
olivia
kim
ophia
nguyen
```

Option 8 Output:

```
No Most Frequent Name
```

## Test Case 2:

Input:

```
Lucas,Emma,Grace kim,Martinez,Chloe Nguyen,Ryan Patel,robinson,Lucas
Garcia,emma,lucas,Mia Brown,Benjamin Thompson,Lily Lucas,Samuel White,Ava
Jones,Emma,Davis
```

Option 1 Output:

```
[Ava Jones, Benjamin Thompson, Chloe Nguyen, Davis, Emma, emma, Emma, Grace
kim, Lily Lucas, Lucas, lucas, Lucas Garcia, Martinez, Mia Brown, robinson,
Ryan Patel, Samuel White]
```

Option 2 Output:

```
Ava Jones
Benjamin Thompson
Chloe Nguyen
Grace kim
Lily Lucas
Lucas Garcia
Mia Brown
Ryan Patel
Samuel White
```

Option 3 Output:

```
Davis
Emma
emma
Emma
Lucas
lucas
Martinez
robinson
```

Option 4 Output:

```
Name Count: 17
Letter Count Total: 134
```

```
Avg Name Length: 7.88
Shortest Name: Emma
Longest Name: Benjamin Thompson
Population Standard Deviation: 3.14
```

Option 5 Output:

```
Ava Jones
Benjamin Thompson
Emma
emma
Emma
Grace kim
Martinez
Mia Brown
robinson
```

Option 6 Output:

```
Chloe Nguyen
Davis
Lily Lucas
Lucas
lucas
Lucas Garcia
Ryan Patel
Samuel White
```

Option 7 Output:

```
emma
kim
lucas
robinson
```

Option 8 Output:

```
Most Frequent Name: Emma
```

## Test Case 3:

Input:

```
Jasper O'Connell,Freeya,Kai johnson,Murphy,Amir Ali,Nia Williams,Kai
Johnson,Tessa Wright,Felix,Lila Patel,kai Johnson,Felix,Kai Johnson,Iris
Taylor,Felix
```

Option 1 Output:

```
[Amir Ali, Felix, Felix, Felix, Freeya, Iris Taylor, Jasper O'Connell, Kai
johnson, Kai Johnson, kai Johnson, Kai Johnson, Lila Patel, Murphy, Nia
Williams, Tessa Wright]
```

Option 2 Output:

```
Amir Ali
Iris Taylor
Jasper O'Connell
Kai johnson
Kai Johnson
kai Johnson
Kai Johnson
Lila Patel
Nia Williams
Tessa Wright
```

Option 3 Output:

```
Felix
Felix
Felix
Freeya
Murphy
```

Option 4 Output:

```
Name Count: 15
Letter Count Total: 130
Avg Name Length: 8.67
Shortest Name: Felix
Longest Name: Jasper O'Connell
Population Standard Deviation: 2.80
```

Option 5 Output:

```
Freeya
Iris Taylor
Kai johnson
Kai Johnson
kai Johnson
Kai Johnson
Murphy
```

Option 6 Output:

```
Amir Ali
Felix
Felix
```

```
Felix
Jasper O'Connell
Lila Patel
Nia Williams
Tessa Wright
```

Option 7 Output:

```
johnson
kai
```

Option 8 Output:

```
Most Frequent Name: Kai johnson
```

## Sample Output:

```
Enter List of Names Separated by Commas:
Emily Johnson,michael,Isabella Martinez,aiden,ophia nguyen,Lucas,olivia
kim,Ethan,ava patel,Noah

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

1
[aiden, ava patel, Emily Johnson, Ethan, Isabella Martinez, Lucas, michael,
Noah, olivia kim, ophia nguyen]


Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program
```

```
2
ava patel
Emily Johnson
Isabella Martinez
olivia kim
ophia nguyen



Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

3
aiden
Ethan
Lucas
michael
Noah

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

4
Name Count: 10
Letter Count Total: 82
Avg Name Length: 8.20
Shortest Name: Noah
```

```
Longest Name: Isabella Martinez
Population Standard Deviation: 3.66

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

5
ava patel
Emily Johnson
Isabella Martinez
Noah

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

6
aiden
Ethan
Lucas
michael
olivia kim
ophia nguyen

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
```

```
9) Enter new list of Names
0) Quit Program

7
aiden
ava
patel
michael
olivia
kim
ophia
nguyen


Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

8
No Most Frequent Name

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

9
Enter List of Names Separated by Commas:
Testing test

Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
```

```
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

1
[Testing test]



Please make a selection:
1) Display List Ordered
2) Display Full Names
3) Display Single Names
4) Display Name Statistics
5) Display Names with Even Length
6) Display Names with Odd Length
7) Display Names not Capitalized
8) Display Most Frequent Name
9) Enter new list of Names
0) Quit Program

0
Program Exiting
```